PARTIAL ASSOCIATION EXPLORER
GLOBAL AND LOCAL (PARTIAL) ASSOCIATIONS IN MIXED-TYPE DATA

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1 - Introduction and Workflow

- **Partial Association Explorer** is an open-source **R Shiny application** for profiling unconditional and conditional associations in multivariate datasets containing **numerical and categorical variables**
- For every selected variable pair, the app computes a global association measure, a significance test, and local diagnostics adapted to the variable types; users can then filter an interactive association network and inspect retained edges through harmonized pair plots

Choose a dataset → Select variables and controls → Compute pairwise associations and tests → Filter the network view → Open local pair plots

2 - Association Measures

Variable types	Unconditional measure	Conditional measure
Numerical–numerical	Pearson's r	Partial r (added-variable regression)
Numerical–categorical	η^2 (ANOVA)	Partial η^2 (ANCOVA)
Categorical–categorical	V_L	$V_{L\mid Z}$

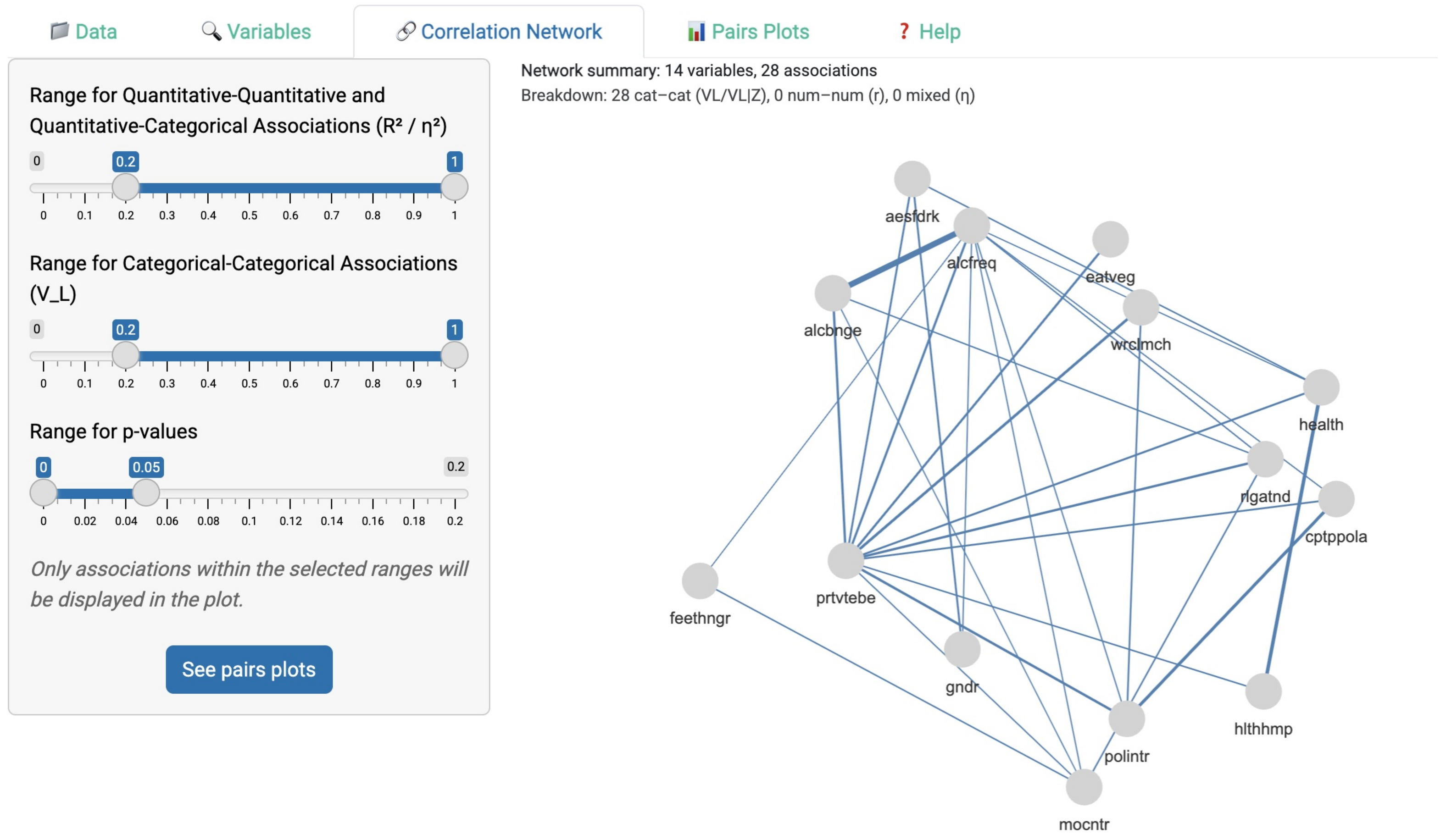
Categorical–categorical specification

$$V_L = \sqrt{1 - \exp\left(-\frac{G^2}{n}\right)}$$

$$G^2 = 2(\ell_1 - \ell_0)$$

Here, ℓ_0 is the fitted log-likelihood under the null model of (conditional) independence, whereas ℓ_1 is the fitted log-likelihood under the alternative model that allows association between the two categorical variables.

3 - Filtered Network

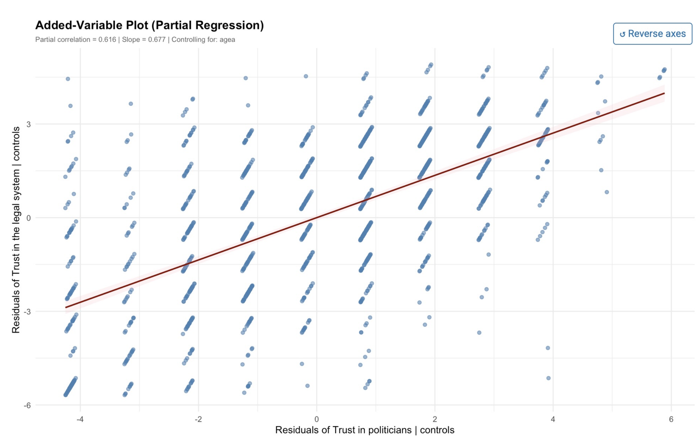


- **Nodes** are selected variables; **edges** are retained pairwise associations
- Range sliders restrict the displayed graph by **p-value** and by the strength of the relevant association measure for each data-type case

4 - Pair Plots

4.1 Numerical–numerical

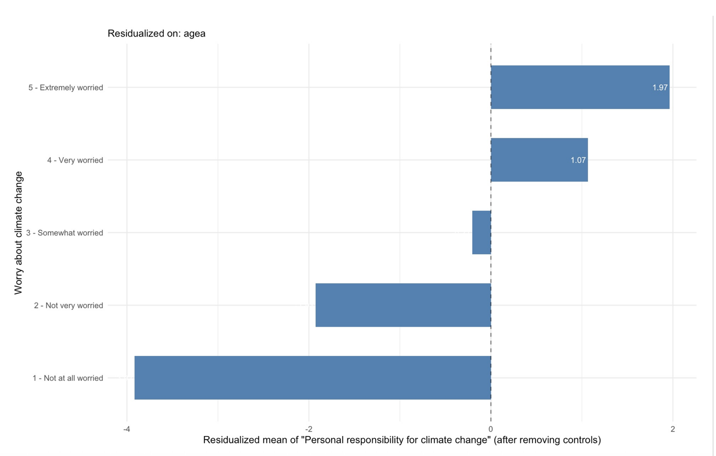
- Scatter plot without controls; added-variable plot with controls



positive slope = positive association; negative slope = negative association

4.2 Numerical–categorical

- Raw group means without controls; residual group means with controls



Stronger separation between categories = stronger association

4.3 Categorical–categorical

- The pair plot is a local association table built from observed counts O_{ij} and expected counts E_{ij} under the null model
- Cell values show $D_{ij} = O_{ij} - E_{ij}$: D_{ij} measures the departure from the null model in count units
- Cell colors are based on $R_{ij} = (O_{ij} - E_{ij})/\sqrt{E_{ij}}$: R_{ij} standardizes this departure relative to its expected variability. A stronger red color indicates that the observed count is substantially higher than expected under independence. A stronger blue color indicates that the observed count is substantially lower than expected under independence.

Unconditional categorical association

VL = 0.405 | p-value = 3.83e-14

Cell values show D = O - E0. Colors are based on Pearson residuals R: red = over-represented, blue = under-represented, darker = stronger.

Large table detected (5x16 = 80 cells). Showing the best 5x7 submatrix selected from likelihood-ratio cell contributions (76.2% of total G^2 contribution).

Worry about climate change	CD&V	Ecolo	Groen!	MR	N-VA	PS	Vlaams Belang
1 - Not at all worried	-1.76	-1.02	-1.65	-3.03	0.12	-1.68	4.29
2 - Not very worried	-4.26	-7.21	-8.77	8.69	2.05	8.10	3.97
3 - Somewhat worried	2.89	-8.95	-17.37	-5.43	16.20	-0.79	16.79
4 - Very worried	5.15	13.18	16.99	4.27	-14.51	-5.32	-18.63
5 - Extremely worried	-2.03	3.99	10.80	-4.51	-3.86	-0.30	-6.41

Local associations between the variables 'Worry about climate change' and 'Party voted for during the last elections'